

Uniform rationality of p -adic factorization densities

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Abstract

Fix a finite extension $K/\mathbb{Q}_p$ with residue field of cardinality q , and let a monic polynomial of degree n over the ring of integers $\mathcal{O}_K$ be drawn at random for the normalized Haar measure on its coefficients. We study the probability $\rho(n, \sigma; q)$ that it factors with a prescribed factorization type σ . We prove that for each n and σ this probability is a single rational function $R_\sigma \in \mathbb{Q}(t)$ evaluated at $t = q$, the *same* function at every prime, including the wild primes $p \mid n$ at which the classical smooth-resolution argument provably breaks down; and that in the projective, weight-normalized normalization the density is palindromic, $R_\sigma(1/t) = R_\sigma(t)$. This settles the rationality and $q \mapsto 1/q$ symmetry conjectured by Bhargava–Cremona–Fisher–Gajić at all primes. The proof is a direct Okutsu–Montes recursion which never constructs a smooth model: the densities are Haar *volumes*, equivalently finite-field point counts, and the wild degeneration that destroys the resolution lives on tangent spaces, where Frobenius is inseparable, while the count lives on points, where Frobenius is bijective. Palindromy at wild primes is then transferred from the tame primes by the identity theorem for rational functions. The argument builds on standard machinery — the higher-order Montes algorithm of Guàrdia–Montes–Nart and the Igusa–Denef p -adic cell decomposition — which we cite and use as such. For $n \leq 3$ the proof is moreover elementary, the wild boxes being empty. The result is accompanied by a companion certificate documenting a Lean 4 formalization (conditional on a minimal box-Haar identification that mathlib cannot yet construct) together with exact symbolic and PARI-oracle cross-checks.

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1 Introduction

Fix a prime p and a finite extension K of the field of p -adic numbers $\mathbb{Q}_p$, with ring of integers $\mathcal{O}_K$, maximal ideal $\mathfrak{m}$, and residue field $\mathbb{F}_q$ of cardinality $q = p^k$. Consider a random monic polynomial

$$h(z) = z^n + c_{n-1}z^{n-1} + \cdots + c_1z + c_0, \quad c_0, \dots, c_{n-1} \in \mathcal{O}_K, \quad (1.1)$$

where the coefficients c_i are drawn independently from the Haar probability measure on $\mathcal{O}_K$ normalized by $\mu(\mathcal{O}_K) = 1$. Over the local field K the polynomial h factors uniquely into irreducible factors, and each irreducible factor h_j of degree $d_j = e_j f_j$ determines a pair (e_j, f_j) , the ramification index and residue degree of the extension of K generated by a root of h_j . The multiset

$$\sigma = \text{type}(h) = \{(e_1, f_1), \dots, (e_r, f_r)\}, \quad \sum_j e_j f_j = n,$$

is the *factorization type* of h . We write $\rho(n, \sigma; q)$ for the probability that a random h as in (1.1) has factorization type σ , normalized as a projective density in the sense made precise in Section 2.1. These are the quantities of the present paper.

The study of such densities goes back to the question of how many roots in K a random p -adic polynomial has, and to the finer question of its full factorization type. Bhargava, Cremona, Fisher, and Gajić [1] computed the distribution of the number of roots of a random p -adic polynomial and observed two striking features of the answer. The first is that the densities are *rational functions of p* : a sample of their tables reads

$$\rho(2, \{(1, 1), (1, 1)\}; q) = \frac{1}{2}, \quad \rho(3, \{(1, 1)^3\}; q) = \frac{(q^2 + 1)^2}{6\Phi}, \quad \rho(3, \{(1, 1), (1, 2)\}; q) = \frac{q^4 + 1}{2\Phi}, \quad (1.2)$$

with $\Phi = \Phi(q) := q^4 + q^3 + q^2 + q + 1$. It is not even obvious a priori that these densities lie in $\mathbb{Q}$, let alone that they are values of a fixed rational function. The second feature is a symmetry: in the appropriate weight-normalized sense the density of a type is invariant under $q \mapsto 1/q$. Both phenomena were captured by a conjecture which, in the form relevant here, we record as follows.

Conjecture 1.1 (Bhargava–Cremona–Fisher–Gajić [1, Conj. 1.2 *et seq.*]). *For each n and each factorization type σ there is a rational function $R_\sigma \in \mathbb{Q}(t)$ such that the density $\rho(n, \sigma; q)$ equals $R_\sigma(q)$ for every prime power q , and, in the projective weight-normalized normalization, $R_\sigma(1/t) = R_\sigma(t)$.*

For *tame* primes — those with $p \nmid n!$, so that the wild part of the ramification is invisible — both halves of Conjecture 1.1 are accessible by classical means. The functional equation is, in this range, an instance of the Igusa–Denef–Meuser $q \mapsto 1/q$ symmetry of p -adic integrals [2], and the rationality can be obtained from an S_n -equivariant geometric resolution of the discriminant locus, as in the companion treatment of tame factorization densities by Del Corso–Dvornicich [3], Yin [4], and the parent paper [5].¹ The difficulty — and the subject of this paper — lies entirely at the *wild* primes $p \mid n$, where the resolution approach *provably* fails (Section 1.2). Our main result proves both halves of Conjecture 1.1 there as well.

¹We use [5] both for Caruso’s distributional results and, by the same key, for the parent Chebotarev density paper on which this manuscript is modeled; cite keys are inherited from that project’s bibliography.

1.1 The main theorem

Theorem 1.2 (Main theorem). *Fix $n \geq 1$ and a factorization type σ with $\sum_j e_j f_j = n$. Then there is a single rational function $R_\sigma \in \mathbb{Q}(t)$, depending only on n and σ and not on the prime, such that for every finite extension $K/\mathbb{Q}_p$ with residue field of cardinality q ,*

$$\rho(n, \sigma; q) = R_\sigma(q),$$

this identity holding at every prime p , including the wild primes $p \mid n$. Moreover, in the projective weight-normalized normalization of Section 2.1,

$$R_\sigma(1/t) = R_\sigma(t).$$

This proves Conjecture 1.1 in full. The argument uses two pieces of standard, refereed machinery, cited and used as such throughout: the higher-order Okutsu–Montes algorithm of Guàrdia–Montes–Nart [6] and the Igusa–Denef p -adic cell decomposition [2, 7]; the tame functional equation [3–5] enters once, to seed the palindromy transfer. For $n \leq 3$ the proof needs none of the higher-order Montes machinery: the wild Newton boxes are empty (every fractional slope of a degree- ≤ 3 cluster has denominator equal to its segment length, so the Newton polygon alone decides ramification), and the argument is elementary.

We separate the two assertions because they have a different character, in the “decoupling” spirit of the parent paper: rationality is *special*, a reflection of the fact that point counts over $\mathbb{F}_q$ are polynomial in q , whereas the functional equation is *general*, a shadow of a Poincaré-type duality. We accordingly prove rationality first (Sections 3–5) and obtain palindromy afterwards (Section 6).

The novelty is the uniformity *at all primes*. The tame range was available by prior work; what is new is that the *same* rational function governs the wild primes, proved by a characteristic-uniform recursion that never requires a smooth resolution. The first genuinely wild instance, the order-2 Artin–Schreier atom at $n = 4$, $p = 2$ (Example 7.3), already engages the higher-order Montes algorithm; it is corroborated by an independent PARI oracle at the wild prime $p = 2$. The densities produced by the recursion are matched symbolically against the Bhargava–Cremona–Fisher–Gajić root-count moments through $n = 6$, including the first deep-wild Montes order-3 type, as a further cross-check (Section 8).

1.2 Significance: why the wild primes are hard, and the mechanism that resolves them

The interest of Theorem 1.2 is that it reaches the wild primes $p \mid n$, where the natural geometric proof of rationality *cannot* be made to work. That proof, used in the tame range, resolves the discriminant slice S_n -equivariantly and reads the density off the resolved model. At a wild prime the slice is non-reduced and there is no S_n -equivariant smooth complement to resolve against. The obstruction is explicit at $(n, p) = (3, 3)$: the cusp of the cubic discriminant acquires a non-reduced tangent cone in characteristic 3, and the equivariant splitting the resolution requires does not exist. This is not a gap to be patched; the method is structurally blind to the wild range. (Indeed μ_n is linearly reductive in every characteristic while S_n is wildly ramified once $p \leq n$, so even stacky destackification, which needs tameness, does not rescue it.)

The mechanism that does work, and the conceptual heart of the paper, is a change of functor.

The densities are Haar volumes, equivalently $\mathbb{F}_q$ -point counts. A volume is invisible to the tangent-space degeneration that breaks the resolution. At a wild prime the relevant geometric loci have characteristic-dependent tangent spaces — Frobenius is inseparable there — but characteristic-independent point counts, because Frobenius is bijective on points. The two facts that broke and that survived are the same fact, read on two different functors.

Resolving a locus reads its tangent space and therefore sees the wild inseparability; measuring it reads only the point count and therefore does not. We compute the density directly, by a recursion that never forms a smooth model. The cleanest instance is already classical: by Serre’s mass formula [8] the totally ramified Eisenstein polynomials of degree n fill a box of char-independent volume $(q-1)/q^{n+1}$ even at wild $p \mid n$, although the corresponding scheme is non-reduced there. The present paper reads the entire Okutsu–Montes descent in this volume-only register.

1.3 The Montes algorithm, and how the recursion uses it

For the expert we now sketch enough of the higher-order Montes/Okutsu–Montes (OM) algorithm to make the proof credible; the precise imported statements are recorded as Hypothesis 2.2. Fix a monic separable $f \in \mathcal{O}_K[x]$ whose factorization type we wish to determine. The algorithm is a finite, p -independent descent.

- (1) *Newton polygon.* Choose a monic lift ϕ of an irreducible factor of $\bar{f} \in \mathbb{F}_q[x]$ and expand f ϕ -adically. The ϕ -Newton polygon of f is the lower convex hull of the resulting valuations. Its sides of negative slope $\lambda = -h/e$ (in lowest terms) record the ramification: a single side of denominator e forces ramification index a multiple of e in the factors it carries. For $n \leq 3$ the polygon already settles everything — each fractional slope has e equal to the side length — which is why those cases need nothing deeper.
- (2) *Residual polynomial.* To each side of slope λ one attaches a *residual polynomial* $R_\lambda(f)$, a monic polynomial over a finite residue extension $\mathbb{F}_r$ with $[\mathbb{F}_r : \mathbb{F}_q] \mid n$. Its degree equals the lattice length of the side. This is the single most important structural fact for us: the residual lives over a *finite* field at every order, so it is always separable when irreducible (finite fields are perfect), and “the residual has a repeated root” is a discriminant condition with a characteristic-independent point count.
- (3) *Dichotomy.* A multiplicity-one factor of $R_\lambda(f)$ is a *leaf*: it certifies an irreducible p -adic factor of f with explicit (e, f) . A factor of multiplicity ≥ 2 is unresolved and *descends*: one builds a higher key polynomial ϕ' refining ϕ and repeats (1)–(3) over $\mathbb{F}_r$, raising the OM order by one. Descent past order r happens precisely when some order- r residual is non-separable — and over a finite field that is exactly when it is non-squarefree.
- (4) *Tree.* The record of the whole descent — the slopes, residual factorizations, and residue extensions at each node — is the *OM tree* of f ; its combinatorial type, with residue data forgotten down to degrees, is the *shape* T . Guàrdia–Montes–Nart prove the descent terminates at an order bounded p -independently in terms of n .

The proof grafts the volume mechanism onto this tree. We stratify coefficient space by OM shape T and compute, for each T , the Haar volume $C_T(q)$ of the cluster of polynomials of that shape. Two facts make C_T a single rational function of q , independent of p :

- each Newton-polygon cell is a lattice box, of Haar volume a char-independent q -power (building block BB1, from Igusa–Denef cell decomposition); and
- each residual count is a point count of a discriminant stratum over a finite field, hence a universal polynomial in q (building block BB3, the “Frobenius is bijective on points” fact).

Inside a single shape the infinitely many descent depths resum as a geometric self-loop, producing the cyclotomic-type denominators Φ one sees in the BCFG tables. Summing C_T over shapes and crossing a p -independent monic-to-projective bridge gives $\rho(n, \sigma; q) = R_\sigma(q)$. The wild prime never enters the counts; it enters only through the slopes and the lifting, which the volume does not resolve. Palindromy is then a one-line transfer: R_σ is a fixed rational function, it is palindromic at the infinitely many tame primes, and an identity of rational functions valid at infinitely many points is an identity.

1.4 Overview of the proof

Work over an unramified base $K = \mathbb{Q}_{p^*}$, so that $v(p) = 1$ always; this pins all of the characteristic dependence to a single normalization constant. The density $\rho(n, \sigma; q)$ is a Haar volume of a factorization-type region in coefficient space. Changing variables from coefficients to roots in the splitting field introduces the Vandermonde Jacobian $\prod_{i < j} |\alpha_i - \alpha_j|$, so that the integrand is exactly that Jacobian: the densities are volumes, not value-weighted integrals. (This distinction is essential: the value-weighted integral $\int |\text{disc}|$ is *not* a uniform rational function — it equals $q/(q+1)$ at odd p but $7/12$ at $p = 2$ — whereas the volume is.)

Stratify root space by the *Okutsu–Montes tree shape* T , the decorated ultrametric type of the roots. The density then decomposes as a finite sum over tree shapes,

$$\rho(n, \sigma; q) = \sum_T C_T(q), \quad (1.3)$$

where $C_T(q)$ is the Haar volume of the cluster of polynomials of shape T realizing σ . The proof has four movements.

- (i) *Per-shape volume is a uniform rational function* (Section 4). By induction on Okutsu–Montes order, each $C_T(q)$ is built from two characteristic-independent ingredients: a Newton-cell box volume, which is a char-independent q -power (the lattice building block, BB1), and a finite-field residual count, which is a universal polynomial in q because finite fields are perfect and every irreducible residual is separable (the residue building block, BB3). The infinitely many descent depths inside a single shape resum, via a geometric self-loop, to a rational function with cyclotomic-type poles.
- (ii) *Assembly* (Section 5). The cluster volumes are combined through a finite linear system over $\mathbb{Q}(q)$ with p -independent coefficients, yielding the monic density as a uniform rational function.
- (iii) *The monic-to-projective bridge* (Section 5). The projective density ρ is the monic recursion read at the reciprocal argument; $q \mapsto 1/q$ swaps the monic chart with the all-roots-at-infinity chart. The monic density is not palindromic, but the $\mathbb{P}^1$ -symmetric projective density is.
- (iv) *Palindromy at wild primes* (Section 6). The functional equation $R_\sigma(1/q) = R_\sigma(q)$ holds at tame primes by the classical tame functional equation. Both sides are now known to be a single rational function of q ; an identity of rational functions holding at infinitely many q is an identity, so it propagates to the wild primes.

The wild characteristic enters only through $v(p) = 1$, a constant; through the internal structure of the splitting field, which the Haar measure does not resolve; and through the smoothness of strata, which the volume ignores. This is why uniformity survives where the resolution fails.

1.5 Relation to previous work

The distribution of the number of K -rational roots of a random p -adic polynomial was computed by Buhler–Goldstein–Moews–Rosenberg [9] and Evans [10] for low degree, by Kulkarni–Lerario and collaborators [11], by Caruso [5], and in full by Bhargava–Cremona–Fisher–Gajić [1], whose rationality and $p \mapsto 1/p$ symmetry we take as the conjecture under study; see also Roy [12]. The finer factorization densities in the *tame* range were studied by Del Corso–Dvornicich [3], by Yin [4], and in the parent Chebotarev density program [5], where the functional equation appears as a Poincaré duality and the densities are computed by an S_n -equivariant resolution. The present paper reuses that tame computation only as an *input* (the tame functional equation, Section 6); its contribution is the wild-prime uniformity, which the resolution method cannot reach.

The p -adic integration we use is the Igusa–Denef–Meuser theory of local zeta functions; we recall the rationality and $s \mapsto -s$ (equivalently $q \mapsto 1/q$) functional-equation paradigm by citation [2, 7], not by re-derivation, following Denef’s expository conventions. The recursion is organized by the higher-order Okutsu–Montes machinery of Guàrdia–Montes–Nart [6], which provides the OM tree, the factorization of a polynomial given its higher Newton polygons, and the termination of the descent. On the arithmetic-statistics side our volumes are the cluster-refined analogues of the mass formulas of Serre [8] and Bhargava [13, 14]; the volume-side shadow of Serre’s mass formula — that totally ramified Eisenstein polynomials of degree n have char-independent volume $(q-1)/q^{n+1}$ even at wild $p \mid n$ — is the simplest instance of the mechanism of this paper. What is genuinely new is the reading of the Okutsu–Montes descent through the two volume-only building blocks, and the consequent uniformity at wild primes.

1.6 Organization

Section 2 fixes conventions, recalls p -adic integration by citation, and gives a self-contained account of Okutsu–Montes trees, recording the Guàrdia–Montes–Nart theorems we use in the precise form we need them. Section 3 isolates the two characteristic-independent counting engines. Section 4 proves that each per-shape cluster volume is a uniform rational function by induction on Okutsu–Montes order. Section 5 assembles the densities and crosses the monic-to-projective bridge. Section 6 transfers palindromy to the wild primes. Section 7 works out small cases in closed form, including the first genuinely wild atom. Section 8 records the provenance of the result and points to the companion certificate and the cross-checks.

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of the parent Chebotarev density paper, whose correct objection to the wild-prime behaviour of the S_n -equivariant resolution prompted the volume-based argument given here.

2 Preliminaries

2.1 Conventions and notation

Throughout, K is a finite extension of $\mathbb{Q}_p$ with valuation v , ring of integers $\mathcal{O}_K$, maximal ideal $\mathfrak{m}$, uniformizer π , and residue field $\mathbb{F}_q$, $q = p^k$. For the proof we work over the *unramified* extension $K = \mathbb{Q}_{p^k}$, where $v(p) = 1$; this is harmless — a totally ramified base only rescales v — and it is exactly what confines the characteristic dependence to the single normalization constant $v(p) = 1$, see Section 1.4. We normalize Haar measure on $\mathcal{O}_K$ by $\mu(\mathcal{O}_K) = 1$, so that a ball of radius q^{-m} has measure q^{-m} and a box $\prod_i \pi^{m_i} \mathcal{O}_K$ has measure $q^{-\sum_i m_i}$. We write $\overline{\mathbb{F}_q}$ for an algebraic closure and Frob for the q -power Frobenius.

A factorization type is a multiset $\sigma = \{(e_j, f_j)\}_j$ of pairs of positive integers with $\sum_j e_j f_j = n$, recording the ramification and residue degrees of the irreducible local factors. We denote by $\alpha(n, \sigma; q)$ the *monic* density — the Haar measure of the set of monic h as in (1.1) of type σ — and by $\rho(n, \sigma; q)$ the *projective* density, the weight-normalized average over binary forms of degree n obtained by symmetrizing over $\mathbb{P}^1$. The load-bearing distinction (Remark 2.1) is that ρ is a Haar *volume* (equivalently a normalized $\mathbb{F}_q$ -point count), not a value-weighted integral.

Remark 2.1 (ρ is a volume, not a value-weighted integral). This is the load-bearing convention. The densities $\rho(n, \sigma; q)$ are volumes of factorization-type regions; they are *not* integrals weighted by $|\text{disc}|$ or any other character. The two are genuinely different functions of q : the value-weighted integral $\int_{\mathcal{O}_K^2} |a^2 - 4b| d\mu$ equals $q/(q+1)$ at odd p but $7/12$ at $p = 2$, so it is not a uniform rational function. The plain volume of the same type region *is* uniform. All statements below concern volumes.

2.2 p -adic integration

We use only the standard p -adic integration formalism, which we recall in outline and otherwise cite. Haar measure on $\mathcal{O}_K^n$ is the product measure, and on a measurable change of variables ϕ with p -adic Jacobian J_ϕ one has $d(\phi_*\mu) = |J_\phi| d\mu$; for the coefficients-to-roots map the Jacobian is the Vandermonde $\prod_{i < j} (\alpha_i - \alpha_j)$. Integrals of the form $\int_X |f|^s d\mu$ are rational functions of q^{-s} and satisfy the Igusa–Denef–Meuser functional equation under $q^{-s} \mapsto q^s$; we recall this paradigm purely by citation [2, 7] and never re-derive it. The single quantitative fact we use repeatedly is that the volume of a Newton-polygon cell is an explicit q -power determined by the cell’s lattice data, which is Building Block BB1 of Section 3.

2.3 Okutsu–Montes trees

The central tool is the Okutsu–Montes (OM) description of the factorization of a p -adic polynomial. We recall it to the extent the proof needs it and use the machinery of Guàrdia–Montes–Nart [6]; theorem numbers below refer to that paper (*Trans. Amer. Math. Soc.* **364** (2012), no. 1, 361–416; arXiv:0807.2620v2).

Given a monic separable $f \in \mathcal{O}_K[x]$ and a monic $\phi \in \mathcal{O}_K[x]$ irreducible mod $\mathfrak{m}$, the ϕ -Newton polygon of f has sides of various slopes, and to each side S of finite slope $\lambda = -h/e$ (in lowest terms, $e \leq n$) one associates a *residual polynomial* $R_\lambda(f) \in \mathbb{F}_r[y]$ over a *finite* residue field $\mathbb{F}_r$, where $[\mathbb{F}_r : \mathbb{F}_q] = f_0 f_1 \cdots | n$. A factor of multiplicity 1 in $R_\lambda(f)$ is a *leaf*: an irreducible p -adic factor with explicit (e, f) . A factor of multiplicity ≥ 2 is unresolved and *descends* to the next OM order, where one repeats the construction over a higher residue field. The data of a polynomial’s full descent — the sequence of slopes, residual factors, and residue extensions — is its *OM tree*, and the *shape* T of the tree is its combinatorial type with the residue-field data forgotten down to degrees. We use the following facts from [6], in the form stated; the theorem numbers are pinned in the certificate’s citation ledger.

Theorem 2.2 (Guàrdia–Montes–Nart [6]). *Let $K/\mathbb{Q}_p$ be a finite extension with finite residue field $\mathbb{F}_q$. For a monic separable $f \in \mathcal{O}_K[x]$:*

- (GMN1) (Finite-field residuals at every order; GMN §2.1, Def. 2.21.) *The order- r residual polynomial $R_\lambda(f)$ is a monic polynomial of explicit degree over a finite field $\mathbb{F}_r$ with $[\mathbb{F}_r : \mathbb{F}_q] = f_0 \cdots f_{r-1} | n$.*

- (GMN2) (Lattice polygons at every order; GMN Def. 2.3, Def. 1.1.) *The order- r Newton polygon N_r is a principal (lattice) polygon, with p -independent combinatorics.*
- (GMN3) (Leaf-or-descend dichotomy; GMN Cor. 1.20, Cor. 3.8, Lem. 3.11.) *A multiplicity-1 residual factor is an irreducible leaf with explicit (e, f) ; a multiplicity- ≥ 2 factor descends; and descent past order r occurs iff some order- r residual is non-separable.*
- (GMN4) (Termination; GMN Thm. 4.18, Cor. 4.19.) *For monic separable f the descent terminates at finite order, bounded p -independently in terms of n .*

The decisive structural point in Theorem 2.2 is (GMN1): the residuals live over *finite* fields at every order. Finite fields are perfect, so every irreducible residual is separable, and the descent trigger of (GMN3) — “non-separable residual” — coincides with “non-squarefree residual”, a *discriminant locus* with a universal point count. There are no inseparable-irreducible exceptions at any order in any characteristic; the wild, inseparable behaviour is the *generic* case of the GMN machinery, not an exception to it. Theorem 2.2 applies separately at each prime p , but in *identical combinatorial form*: the slopes, residual degrees, residue extensions, and the leaf/descend dichotomy are all expressed in p -independent lattice and finite-field data. The uniformity of the densities is *derived* from this, not assumed.

2.4 The inputs we use

For convenient reference we collect the standard machinery the proof builds on. All three are published, refereed results, cited and used here exactly as one cites any prior theorem.

- (i) the higher-order Okutsu–Montes algorithm of Guàrdia–Montes–Nart, in the form of Theorem 2.2 [6];
- (ii) the Igusa–Denef p -adic cell decomposition and the attendant box-wise measure facts — ball and box volumes, and the per-cell residual fibration — in the form recalled in Sections 2.2 and 3 [2, 7];
- (iii) the tame functional equation: at tame primes $p \nmid n!$ the projective density admits a palindromic rational representative, supplied by the Igusa–Denef–Meuser symmetry and the tame factorization-density literature [2–5]. This is used only once, in Section 6, to seed the palindromy transfer.

3 Residue and cell counts

This section isolates the two characteristic-independent counting engines. They are the precise content of “rationality is special”: both are point counts over $\mathbb{F}_q$, hence polynomials in q with p -independent coefficients.

Lemma 3.1 (BB1: Newton-cell box volume). *Let C be a Newton-polygon cell cut out by valuation conditions on the coefficients, with lattice data (V, A) (V the number of free residual directions, A the lattice height). Then the Haar volume of the corresponding box of polynomials is*

$$\text{vol}(C; q) = (1 - q^{-1})^V q^{-A},$$

a char-independent q -power times a unit factor. In particular it depends only on the combinatorial cell, not on p .

Proof sketch. A Newton-polygon condition is a finite intersection of valuation conditions on the c_i , each of the form $v(c_i) = m_i$ or $v(c_i) \geq m_i$. The measure of such a box is a product of factors q^{-m_i} (for an equality, weighted by the unit-coset factor $1 - q^{-1}$) and q^{-m_i} (for an inequality), by the normalization of Section 2.1. Collecting exponents gives the lattice height A and the number V of unit-coset factors. The exponent is a function of the lattice geometry of the cell alone; see [2] for the cell-decomposition formalism. The explicit lattice exponent at arbitrary OM order is recorded in the certificate. $\square$

Lemma 3.2 (BB3: residual count over a finite field). *Let $\mathbb{F}$ be a finite field with $|\mathbb{F}| = Q$. The number of monic non-squarefree polynomials of degree $d \geq 2$ over $\mathbb{F}$ is Q^{d-1} , independently of the characteristic. Equivalently, the locus of residual polynomials that fail to be separable — the locus that triggers OM descent by Hypothesis 2.2(GMN3) — is the discriminant locus, of universal point count Q^{d-1} , with no inseparable-irreducible exceptions in any characteristic.*

Proof. The number of monic squarefree polynomials of degree $d \geq 2$ over $\mathbb{F}$ is $Q^d - Q^{d-1}$; hence the non-squarefree count is Q^{d-1} . Over a finite field every irreducible polynomial is separable, because $\mathbb{F}$ is perfect: every element is a p -th power, so $y^p - c = (y - c^{1/p})^p$ is never irreducible. Thus “non-separable” and “non-squarefree” coincide, and the descent-triggering locus is the discriminant locus with the stated universal count. $\square$

Lemma 3.2 is the precise meaning of “Frobenius is bijective on points”. The wild inseparability lives entirely in the p -adic tree structure — the slopes and the lifting — and never in the residue-field counts, which are always the universal discriminant-stratum polynomials. This is what makes the recursion characteristic-uniform.

4 The cluster recursion

We now prove that each per-shape cluster volume is a uniform rational function of q . The argument is an induction on Okutsu–Montes order, resting only on Lemma 3.1 and Lemma 3.2.

4.1 Splitting and the Vandermonde pushforward

If $\bar{f} = \bar{g}\bar{h}$ with $\bar{g}, \bar{h}$ coprime mod $\mathfrak{m}$, then Hensel’s lemma identifies $f \mapsto (g, h)$ with a measure-preserving product bijection: the resultant $|\text{Res}(g, h)| = 1$ is a unit, so the conditional law of a cluster’s translated coefficients is Haar on the corresponding congruence cell. This is the exact product structure that lets the recursion factor through coprime clusters. On a fixed cluster, changing variables from coefficients to roots in the splitting field introduces the Vandermonde Jacobian; on a stratum of fixed tree shape T the pairwise valuations $v(\alpha_i - \alpha_j)$ are the tree distances and are constant, so

$$\prod_{i < j} |\alpha_i - \alpha_j| = q^{-d(T)}$$

is a single q -power determined by the shape. The pushforward of Haar measure over the stratum is then the residue-configuration count times $q^{-(\text{cell codimension})}$, both explicit by Section 3.

4.2 Per-shape volume by induction on OM order

Proposition 4.1 (Per-shape uniform rationality). *For each OM tree shape T on n roots, the cluster volume $C_T(q)$ is a uniform rational function of q , the same at every prime including wild $p \mid n$.*

Proof outline. Induct on the OM order r of the shape. At order r , the cluster is partitioned by the order- r Newton polygon into cells; by Hypothesis 2.2(GMN2) these are lattice cells, so each has box volume a char-independent q -power by Lemma 3.1. Within a cell, the order- r residual polynomial lives over the finite field $\mathbb{F}_r$ by Hypothesis 2.2(GMN1); a multiplicity-1 residual factor is a leaf with explicit (e, f) by (GMN3), contributing a $\mathbb{F}_r$ -point count that is a universal polynomial in q by Lemma 3.2, and a multiplicity- ≥ 2 factor descends to a strictly smaller child shape, to which the inductive hypothesis applies. The passage from this finite-field *count* to a Haar *volume* at order r is the order- r analogue of Lemma 3.1: by Theorem 2.2(GMN1)–(GMN2) the order- r residual coordinates are free $\mathbb{F}_r$ -digits over a lattice cell, so the per-cell volume is the residual point count times the char-independent box volume of Lemma 3.1, exactly as at order 1. (At order 1 this is elementary; the order- r form is a direct consequence of the Guàrdia–Montes–Nart coordinate facts and the Igusa–Denef cell decomposition, and is independently oracle-corroborated through order 4; see Section 8.) The number of cells, the slopes, and the residue degrees are all p -independent, and the descent terminates at finite p -independent order by (GMN4). Finally, a single shape contains infinitely many descent depths — the totally interior rescaling $\{v(c_i) \geq i\}$ loops the recursion back to the global problem at a rescaled argument — and these resum as a geometric series with ratio a q -power, contributing a denominator $1 - q^{-(e(e+1)/2-1)}$ for a cluster of size e (for $e = 3$ this is $q^5 - 1$ up to a unit, the classical Φ -type denominator). The result is a rational function of q with cyclotomic-type poles, assembled from p -independent data; it is therefore the same rational function at every prime. $\square$

Remark 4.2 (Where the wildness goes). The only place a wild prime could intervene is in the residual polynomials at a descent node, where in characteristic p a repeated factor is inseparable (Artin–Schreier). By Lemma 3.2 this is invisible to the count: the non-separable locus is the discriminant

locus with the universal count Q^{d-1} , the same geometric data as the tame “repeated root” locus at an odd prime, even though the two loci are different subvarieties with different tangent cones. This is the volume-side image of the referee’s tangent-space obstruction: the loci that the resolution cannot smooth uniformly have nonetheless the same volume in every characteristic.

5 Assembling the densities

5.1 From cluster volumes to the monic density

By the tree decomposition (1.3) and Proposition 4.1, the monic density $\alpha(n, \sigma; q)$ is a finite sum of cluster volumes, each a uniform rational function of q . Multi-cluster types are linearized: the coefficients relating the type densities to the cluster volumes form a finite linear system over $\mathbb{Q}(q)$ whose entries are residue-configuration counts and q -powers, all p -independent. Solving the system expresses $\alpha(n, \sigma; q)$ as a uniform rational function of q .

5.2 The monic-to-projective bridge

Both densities are the same stratification of binary degree- n forms by their residue divisor on $\mathbb{P}^1$, driven by the same intrinsic cluster laws L_e :

$$\rho(n, \sigma; q) = \frac{1}{|\mathbb{P}^n(\mathbb{F}_q)|} \sum_{D \in \text{Sym}^n \mathbb{P}^1(\mathbb{F}_q)} W_D[\sigma], \quad \alpha(n, \sigma; q) = \frac{1}{q^n} \sum_{D \in \text{Sym}^n \mathbb{A}^1(\mathbb{F}_q)} W_D[\sigma],$$

with the same weights W_D built from the cluster laws. The only difference is whether the point $[1 : 0] \in \mathbb{P}^1$ is admissible, i.e. $q+1$ projective versus q affine base points, together with the normalization $|\mathbb{P}^n(\mathbb{F}_q)|$ versus q^n ; the crux identity is $\text{Sym}^n \mathbb{P}^1 = \mathbb{P}^n$. Consequently the projective density is the monic recursion read at the *reciprocal* argument. This is why the monic density is not palindromic — indeed the monic $\alpha(3, \{(1, 1)^3\}; 2)$ is $4/93$, not symmetric under $q \mapsto 1/q$ — while the $\mathbb{P}^1$ -symmetric projective ρ is palindromic. The bridge is itself p -independent: it is built from point counts and is the volume-side incarnation of the symmetry-for- α identity $\beta(e; q^\delta) = \alpha(e; q^{-\delta})$.

Corollary 5.1 (Uniform rationality). *For each n and σ , the projective density $\rho(n, \sigma; q)$ is a uniform rational function $R_\sigma(q) \in \mathbb{Q}(t)$, the same at every prime including wild $p \mid n$. This is the rationality half of Theorem 1.2.*

6 The functional equation at wild primes

It remains to prove palindromy, $R_\sigma(1/t) = R_\sigma(t)$. The argument is the clean payoff of the decoupling: we do not prove the functional equation at wild primes directly; we transfer it.

Proposition 6.1 (Palindromy). *The projective density satisfies $R_\sigma(1/t) = R_\sigma(t)$.*

Proof. By Corollary 5.1 a single rational function $R_\sigma \in \mathbb{Q}(t)$ computes the projective density at *every* prime: $R_\sigma(q) = \rho(n, \sigma; q)$ for each prime power q , tame or wild. At a tame prime $p \nmid n!$ the tame functional equation (the Igusa–Denef–Meuser symmetry and the tame factorization-density literature, input (iii) of Section 2.4) furnishes a *palindromic* rational function $P_\sigma \in \mathbb{Q}(t)$, $P_\sigma(1/t) = P_\sigma(t)$, that also computes the density there: $P_\sigma(q) = \rho(n, \sigma; q)$ for all but finitely many tame prime powers q . Thus R_σ and P_σ agree at infinitely many values — for any fixed n there are infinitely many tame primes, with unboundedly large residue cardinalities — so the polynomial numerator of $R_\sigma - P_\sigma$ has infinitely many roots and vanishes: $R_\sigma = P_\sigma$ in $\mathbb{Q}(t)$. Therefore $R_\sigma(1/t) = P_\sigma(1/t) = P_\sigma(t) = R_\sigma(t)$ identically, in particular at the wild primes $p \mid n$, where the tame functional equation itself is unavailable. $\square$

Together, Corollary 5.1 and Proposition 6.1 prove Theorem 1.2.

7 Examples

We record the small cases in closed form, with $\Phi = \Phi(q) = q^4 + q^3 + q^2 + q + 1$. The densities below are projective, weight-normalized, and exact rational functions of q ; palindromy and the sum-to-one normalization are visible by inspection.

Example 7.1 (Quadratics, $n = 2$). The three types are split $\{(1, 1), (1, 1)\}$, inert $\{(1, 2)\}$, and ramified $\{(2, 1)\}$:

$$\rho(2, \{(1, 1)^2\}; q) = \frac{1}{2}, \quad \rho(2, \{(1, 2)\}; q) = \frac{q^2 - q + 1}{2(q^2 + q + 1)}, \quad \rho(2, \{(2, 1)\}; q) = \frac{q}{q^2 + q + 1}.$$

Each is invariant under $q \mapsto 1/q$ and the three sum to 1. The ramified type includes the wild prime $p = 2$ with no exception. As an independent check, a direct 2-adic computation at the wild prime gives the *monic* densities $\alpha(2, \{(1, 1)^2\}, \{(1, 2)\}, \{(2, 1)\}; 2) = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, exactly the values at $q = 2$ of the uniform rational functions assembled by the recursion — the wild prime $p = 2$ obeys the same formula as every tame prime, with no exception. This is genuinely subtle at $p = 2$: a monic quadratic whose reduction is an inseparable square, $x^2 + ax + b \equiv (x + c)^2 \pmod{2}$, may be split, inert, or ramified, so the reduction no longer determines the type (Hensel's lemma fails at the double root), yet the three densities still land on the uniform values. The *projective* densities ρ at $q = 2$, read off the formulas above, are instead $(\frac{1}{2}, \frac{3}{14}, \frac{2}{7})$: the monic densities are uniform but not palindromic, whereas the $\mathbb{P}^1$ -symmetric projective densities are both, the monic-to-projective bridge of Section 5.2 accounting for the difference.

Example 7.2 (Cubics, $n = 3$). The five types are $\{(1, 1)^3\}$, $\{(1, 1), (1, 2)\}$, $\{(1, 3)\}$, the partially ramified $\{(1, 1), (2, 1)\}$, and the totally ramified $\{(3, 1)\}$:

$$\begin{aligned} \rho(3, \{(1, 1)^3\}; q) &= \frac{(q^2 + 1)^2}{6\Phi}, & \rho(3, \{(1, 1), (1, 2)\}; q) &= \frac{q^4 + 1}{2\Phi}, \\ \rho(3, \{(1, 3)\}; q) &= \frac{q^4 - q^2 + 1}{3\Phi}, & \rho(3, \{(1, 1), (2, 1)\}; q) &= \frac{q^3 + q}{\Phi}, \\ \rho(3, \{(3, 1)\}; q) &= \frac{q^2}{\Phi}. \end{aligned}$$

All five are palindromic under $q \mapsto 1/q$ and sum to 1. The totally ramified type $\{(3, 1)\}$ includes the wild prime $p = 3$ with no exception — precisely the case $(n, p) = (3, 3)$ at which the S_3 -equivariant resolution fails, yet the density is the same uniform rational function q^2/Φ .

Example 7.3 (The first wild atom, $n = 4$, $p = 2$). The first genuinely wild phenomenon appears in a size-4 cluster with a single Newton slope $\frac{1}{2}$. Conditioning on a clean slope- $\frac{1}{2}$ cluster, the residual polynomial is a quadratic $R(y) = y^2 + \bar{b}y + \bar{a}$ with $\bar{a}$ uniform in $\mathbb{F}_q^\times$ and $\bar{b}$ uniform in $\mathbb{F}_q$. The descent-triggering locus — R has a repeated root — is

$$\{\bar{b}^2 = 4\bar{a}\} \text{ at odd } p \quad \text{versus} \quad \{\bar{b} = 0\}, \quad R = (y + \sqrt{\bar{a}})^2 \text{ at } p = 2,$$

which is *inseparable* at $p = 2$ (Artin–Schreier). These are geometrically different loci — a punctured parabola versus a punctured line, with different tangent cones — but both are the image of the merge map $\rho \mapsto (y - \rho)^2$ and so have the *same* measure $1/q$ in every characteristic, by Lemma 3.2. Resolving the full slope- $\frac{1}{2}$ cluster gives the characteristic-uniform conditional cluster law

$$P((4, 1) \mid \text{inseparable sub-stratum}) = \frac{q}{q + 1},$$

the wild $p = 2$ degeneration yielding the *same* $q/(q + 1)$ as the tame double point at an odd prime. This is the deepest instance of the mechanism: the char-dependent geometry is invisible to the volume.

Remark 7.4 (A deep-wild value, $n = 6$). The recursion reaches the first deep-wild Okutsu–Montes order-3 branch at $n = 6$: a slope- $\frac{1}{2}$ cluster whose order-1 residual is a cube $(y - \rho)^3$, forcing a size-3 inseparable child at $p = 2$. The conditional cluster law that the recursion derives for the totally ramified outcome $(6, 1)$ inside this branch is

$$\beta_{(6,1)}(q) = \frac{q^2(q - 1)(q^2 + 1)}{q^5 - 1}, \quad \beta_{(6,1)}(2) = \frac{20}{31},$$

the same rational function at $p = 2$ and at the odd primes. This value $20/31 \approx 0.645$ is what a PARI **factorpadic** oracle confirms at every prime tested, and it *rejects* the naive guess $q/(q + 1) = 2/3$ (rejected at roughly 11σ on a 5-event χ^2). We stress that $\beta_{(6,1)}$ is a *conditional cluster volume*, not

the full projective type density $\rho(6, \{(6, 1)\}; q)$; the latter is a more complicated rational function that is itself palindromic. The cross-check is evidence for the recursion, not a proof; see Section 8.

8 Provenance and verification

Inputs and scope.

Theorem 1.2 is proved using two pieces of standard, published machinery, cited throughout: the higher-order Okutsu–Montes algorithm of Guàrdia–Montes–Nart [6] and the Igusa–Denef p -adic cell decomposition [2, 7]; the tame functional equation [3–5] enters once, to seed the palindromy transfer of Section 6. For $n \leq 3$ the proof is elementary and uses none of the higher-order Montes machinery: the wild Newton boxes are empty (the Newton polygon alone decides the factorization, with no order- ≥ 2 residual invoked). The higher-order algorithm first engages at the order-2 Artin–Schreier atom of Example 7.3 ($n = 4$, $p = 2$), where the argument is additionally corroborated by an independent PARI oracle at the wild prime $p = 2$. We note for the record that the r -point pole structure and related finer questions form a separate line of work, currently partial, and are not treated in this paper.

Cross-checks (evidence, not proof).

The densities produced by the recursion have been compared, as exact rational functions of q , against the Bhargava–Cremona–Fisher–Gajić root-count moments. The full Okutsu–Montes density engine reproduces the BCFG root-count distribution $P(n, r; q)$ and all of its raw and factorial moments *exactly* (symbolic identity, `simplify` = 0) for every degree it reaches, $n \leq 6$, including the deep-wild order-3 type of Remark 7.4, with the exhaustive-enumeration gate $\sum_{\sigma} \rho(n, \sigma; q) = 1$ holding symbolically. A PARI `factorpadic` oracle confirms the individual type densities, with no unpredicted types, at primes 2, 3, 5, 7 within Monte-Carlo noise. These corroborate the result but are not part of the proof. The one step worth singling out is the order- r count-to-volume passage of Proposition 4.1 (the residual-equidistribution step, “M6” in the working notes): it is elementary at order 1 and follows at every order from the Guàrdia–Montes–Nart coordinate facts together with the Igusa–Denef cell decomposition. We have additionally checked it directly against the oracle through OM order 4.

The companion Lean formalization.

A companion certificate accompanies this paper, documenting a Lean 4 / `mathlib` formalization of the count-native path; we record here precisely what it establishes. The algebraic skeleton of the proof is fully machine-checked: the decomposition $\rho = \sum_T C_T$, the rationality of each cluster volume C_T , the geometric self-loop collapse, the finite-field residual count, the box volume, and the palindromy transfer are all genuine Lean theorems whose `#print axioms` footprints are the three Lean-core axioms (`propext`, `Classical.choice`, `Quot.sound`) only. The capstone `goal_theorem_montes` assembles these into the statement that the genuine counting density is a uniform rational and palindromic function of q , with footprint `{propext, Classical.choice, Quot.sound, tame_functionalEquation}`, conditional on a minimal bundle of hypotheses carried as structure fields (not as additional axioms): the Guàrdia–Montes–Nart count-shadow facts and a single box-Haar identification.

The one genuine limitation is specific to the *formalization*, not to the mathematics: `mathlib` has no p -adic Haar measure, so the identification of the per-box p -adic volume with the separately proved lattice/finite-field value cannot yet be constructed inside Lean and is carried as a hypothesis (the “measure wall”). For this reason the formalization should be described as *verified modulo the box-Haar identification*, rather than “formally verified” without qualification. This is a statement about what `mathlib` can currently express; the mathematical theorem itself is proved in the body of this paper.

Byline and collaboration.

The byline of this manuscript is an editorial decision of the project lead, A. G. It credits Claude (Anthropic) as primary author of the present draft because the proof architecture, the structural mechanism of Section 1.2, the worked examples, and the prose were produced in an AI-assisted development loop. A. G. contributed in a human advisory and verification-discipline role: setting the research program, supplying the framing of the inputs used (Section 2.4), running and adjudicating the independent symbolic and PARI-oracle cross-checks above, and enforcing the honesty discipline of this section — in particular that the cited machinery be named, that cross-checks be labelled

evidence rather than proof, and that the companion formalization be described modulo the box-Haar identification it carries. We record this division of labour transparently so that the provenance of every claim is clear.

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